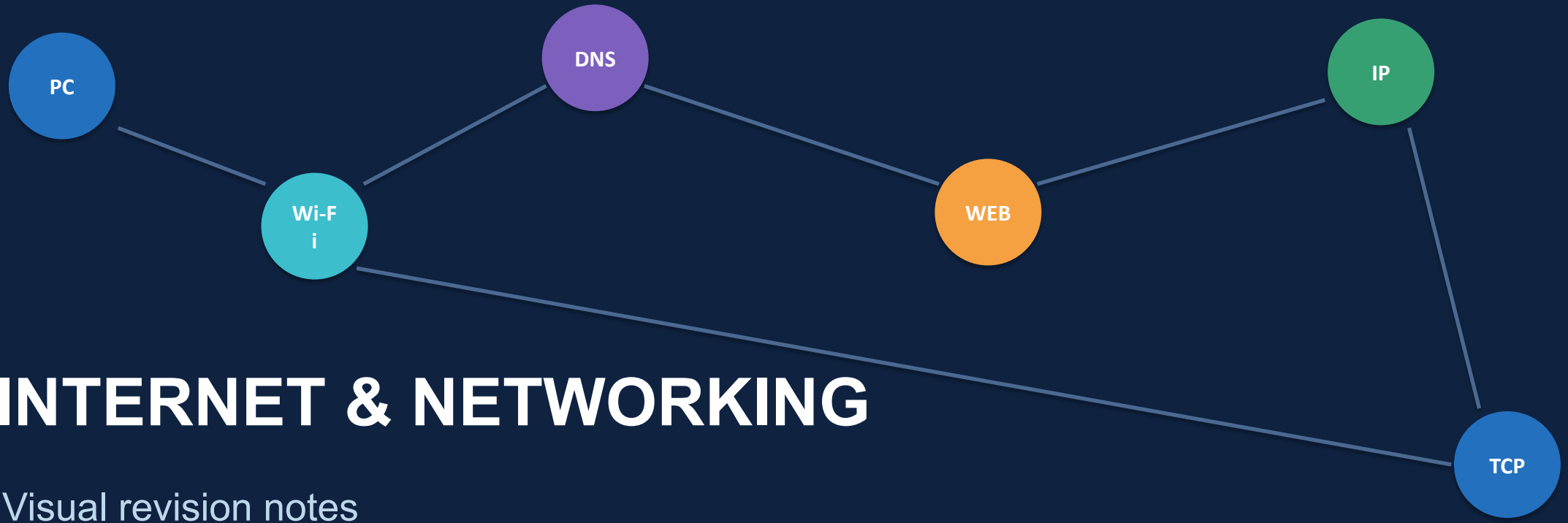


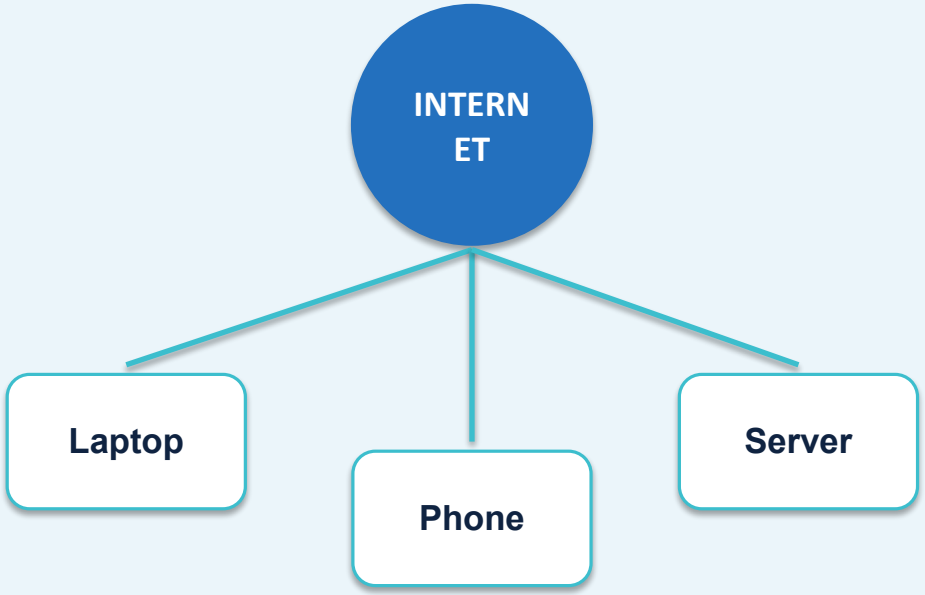
INTERNET & NETWORKING

Visual revision notes

Internet • Networks • Devices • IP • DNS • Protocols

1. Internet vs World Wide Web

Two related ideas — but they are not the same



Internet

Global network infrastructure that connects devices.



Web pages and websites accessed through browsers

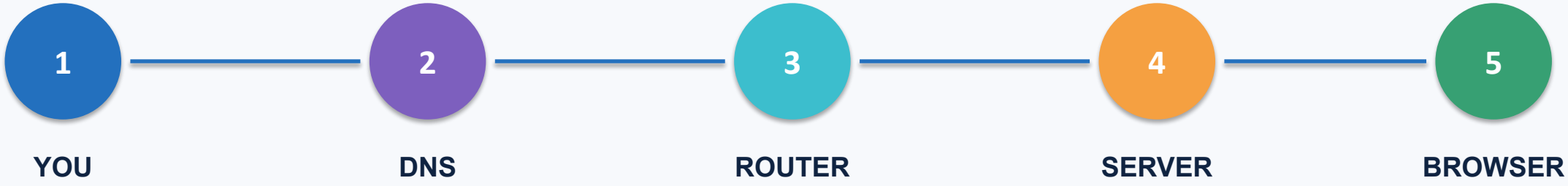
WWW

A collection of linked web pages and resources.

Example: The Internet is the road system; the Web is one service travelling on it.

2. How the Internet works

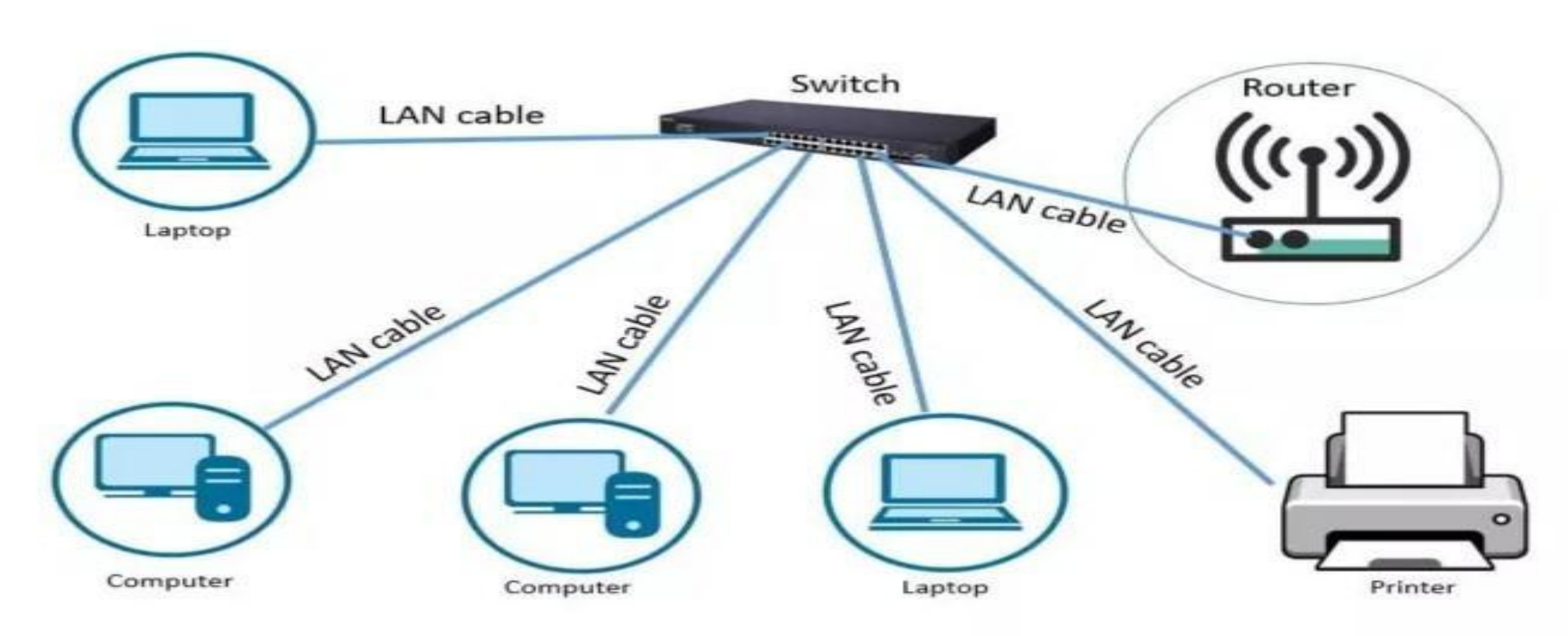
Your device sends a request; a remote server sends back a response



Example: Type www.wikipedia.org → DNS finds an IP → your browser requests the page → the server responds.

3. What exactly is a network?

A network connects two or more devices so they can communicate and share resources



Share resources

Files, printers and storage.

Communicate

Email, chat and video calls.

Access services

Websites, cloud apps and IoT.

4. Types of networks — PAN, LAN, MAN & WAN

The main difference is the geographical area covered



Personal Area Network

A few metres

Example
Phone ↔ Bluetooth earbuds

PERSONAL



Local Area Network

Room / building

Example
Computers in a college lab

LOCAL



Metropolitan Area Network

Across a city

Example
Branches connected within Bengaluru

CITY



Wide Area Network

Countries / world

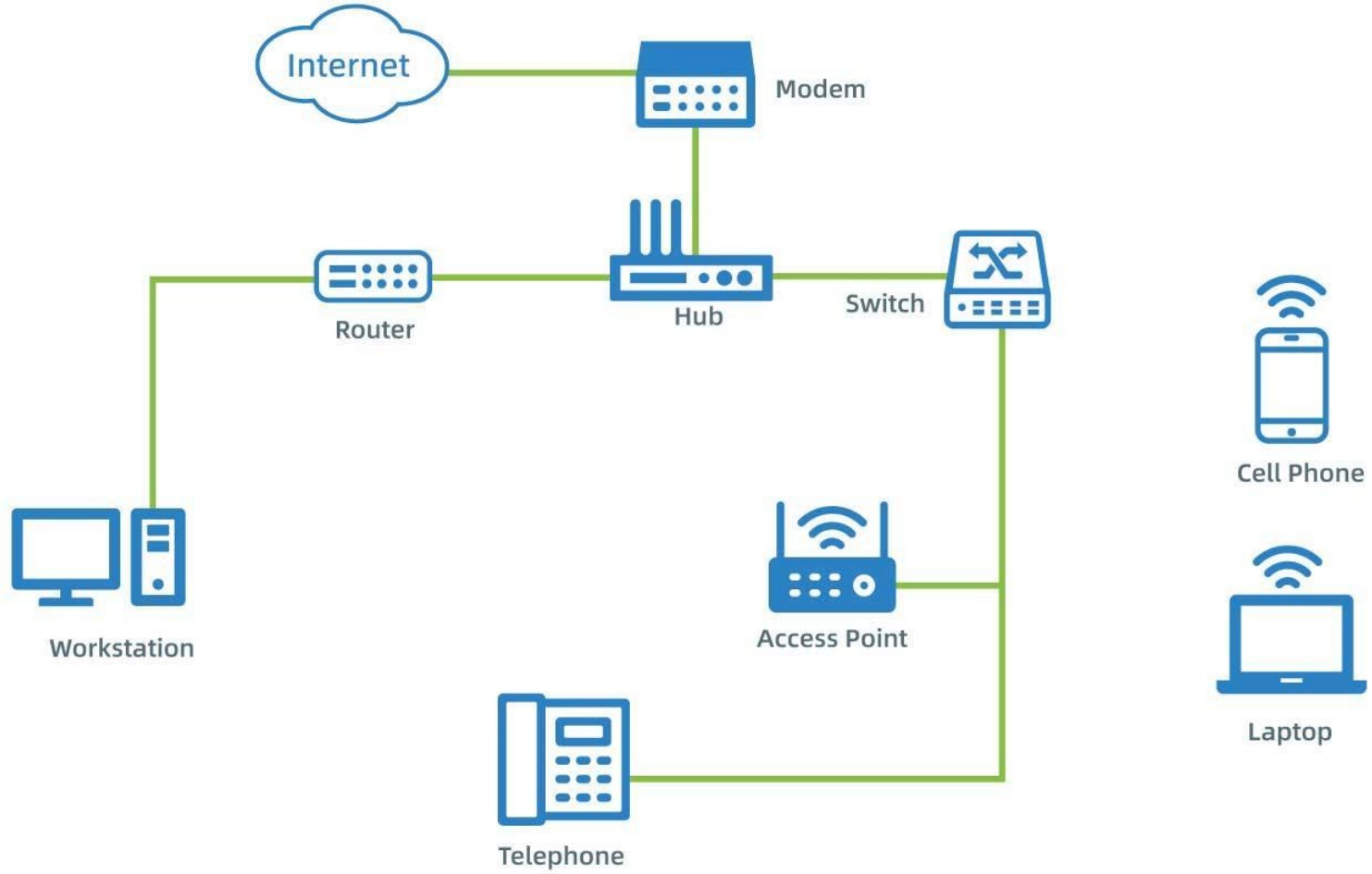
Example
Company offices worldwide; Internet

GLOBAL

Remember: A LAN can work inside a college even when it has no Internet connection.

5. Networking devices — how do computers connect?

Each device has a different job in moving data



5. Networking devices — how do computers connect?

Each device has a different job in moving data

RO

ROUTER

Connects different networks; forwards packets.

Example: Home router connects your LAN to the Internet.

SW

SWITCH

Connects devices inside one local network.

Example: Lab PCs and printer connect to a switch.

MO

MODEM

Connects your network to the ISP access line; converts signal technology.

Example: Fibre/cable/DSL service at home.

AC

ACCESS POINT

Provides Wi-Fi and connects wireless devices to a network.

Example: Students connect laptops to campus Wi-Fi.

ET

ETHERNET CABLE

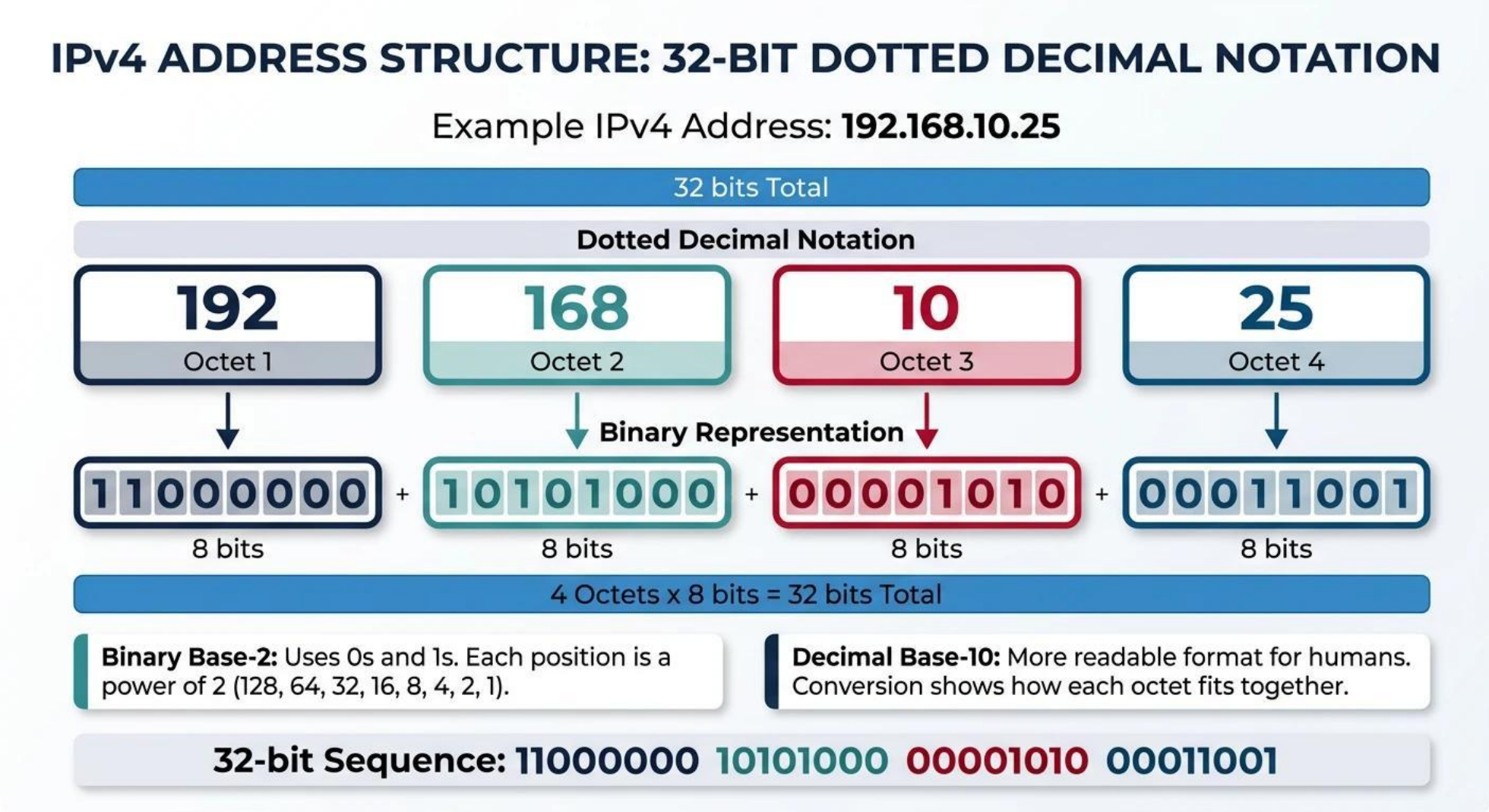
Physical cable carrying network data.

Example: Desktop PC wired to a switch.

Typical path: Device → Switch / Access Point → Router → Modem / ISP → Internet

6. IP address — the network address

An IP address helps identify the source and destination of data packets



6. IP address — the network address

An IP address helps identify the source and destination of data packets

POSTAL ADDRESS ANALOGY



Like a house address helps a courier find a home, an IP address helps data reach the correct device.

PRIVATE IP

PUBLIC IP

Example private IPv4: 192.168.1.10

TWO IP VERSIONS

IPv4 • 32 bits

192.168.1.10
Four decimal numbers separated by dots.

IPv6 • 128 bits

2001:db8::1
Hexadecimal groups separated by colons.

IPv6 provides a much larger address space.

7. DNS — the Internet's name finder

DNS = Domain Name System • It helps find IP information for a domain name

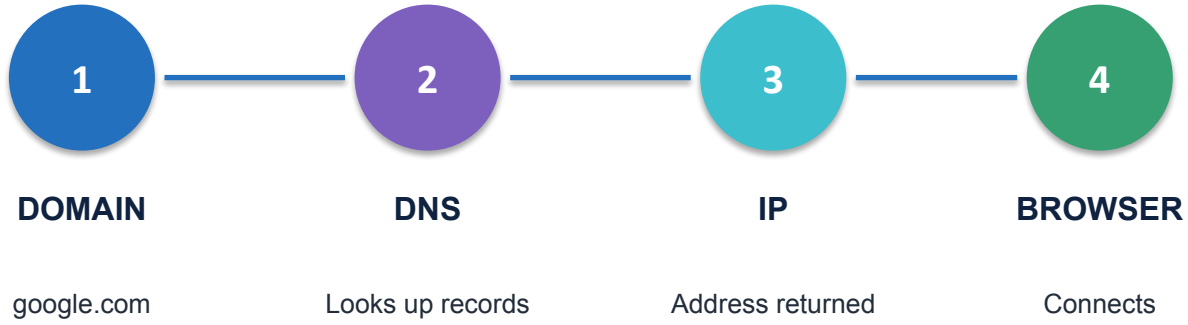
PHONE CONTACTS

Search: Alex

Alex → 98765...

You remember the name, not the number.

DNS LOOKUP



Try in Command Prompt / Terminal: nslookup google.com

8. Understand DNS using a command

A quick practical activity students can run

COMMAND PROMPT / TERMINAL

```
> nslookup google.com

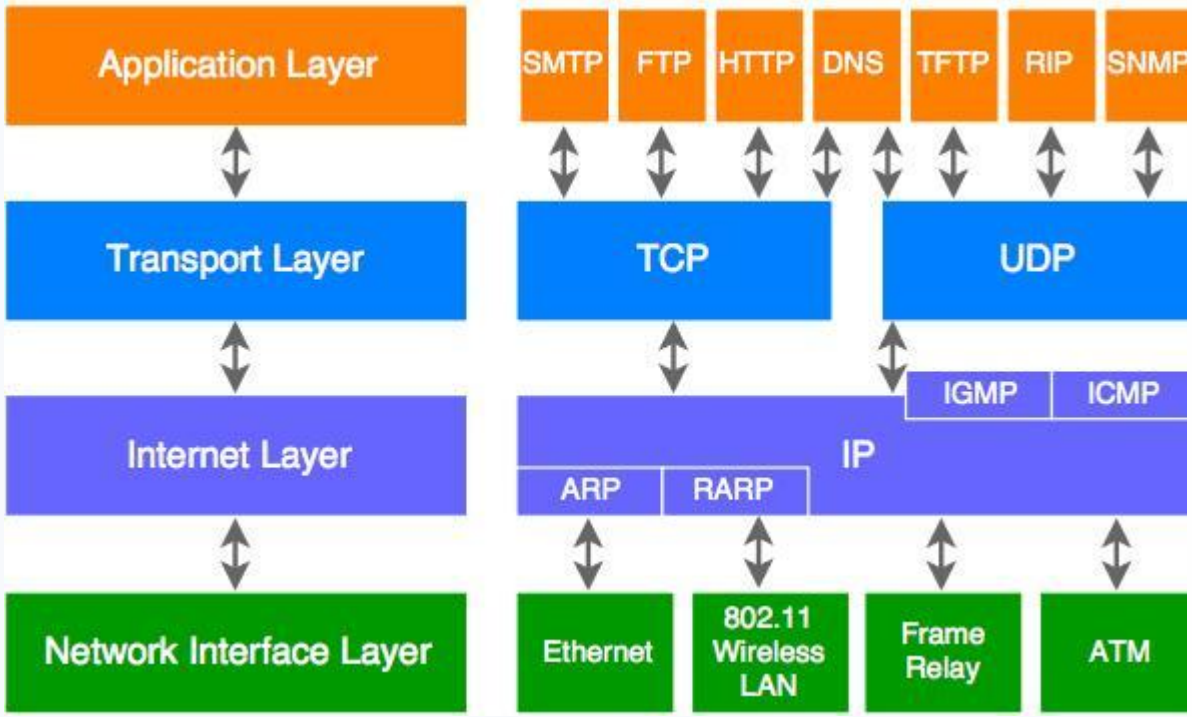
> nslookup youtube.com

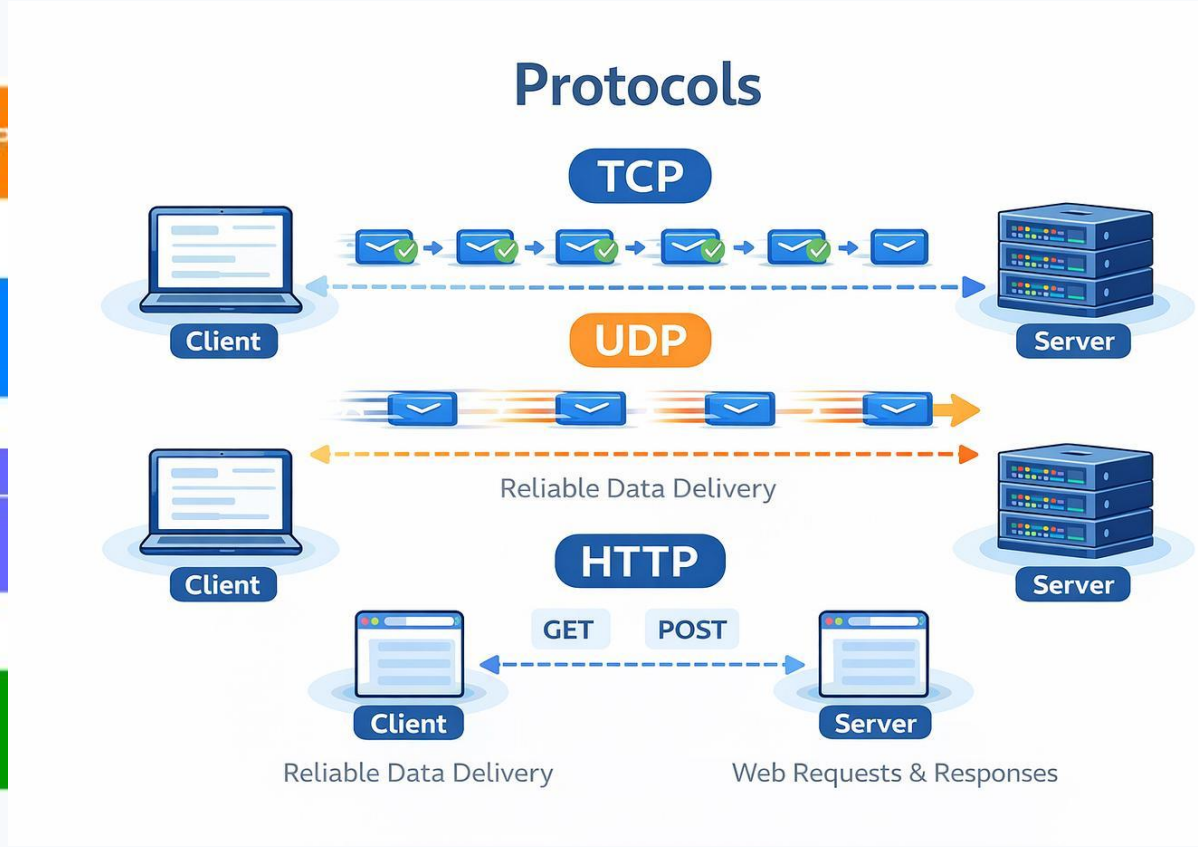
> nslookup wikipedia.org
```

The resolver returns DNS information.
Results can vary by network and time.

9. Protocols — how do devices communicate?

A protocol is an agreed set of rules for exchanging data





9. Protocols — how do devices communicate?

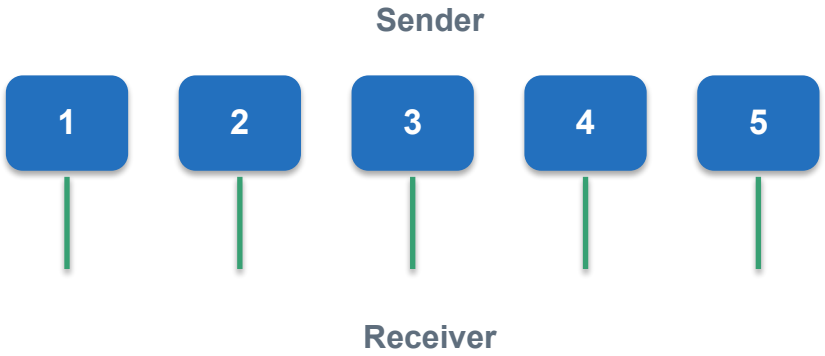
A protocol is an agreed set of rules for exchanging data

H	**HTTP**	Web pages and resources	Example: Browser requests a page	**Port 80**
H	**HTTPS**	Secure web communication using TLS	Example: Online banking website	**Port 443**
F	**FTP**	File transfer between systems	Example: Upload files to a server	**Port 20 / 21**
S	**SMTP**	Sending email between clients/servers	Example: Send a college email	**Port 25***
D	**DNS**	Finds records for domain names	Example: google.com → IP information	**Port 53**
T	**TCP**	Reliable, ordered transport	Example: Web traffic (HTTP/1.1, HTTP/2)	**Port —**
U	**UDP**	Low-overhead datagram transport	Example: Voice/video and games; DNS often	**Port —**

10. TCP vs UDP

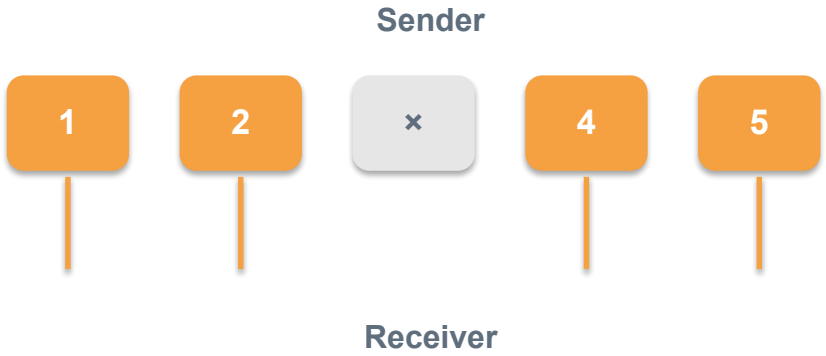
Both are transport protocols — they make different trade-offs

TCP • RELIABLE & ORDERED



Acknowledgements + retransmission help deliver data in order.
Analogy: tracked courier package.

UDP • LIGHTWEIGHT & FAST



No delivery/order guarantee at UDP layer; keeps going.
Analogy: live announcement.

QUICK REVISION

Six questions to remember

INTERNET
What connects devices globally?

IP
Where should the data go?

DNS
Which IP information belongs to this domain?

HTTP / HTTPS
How do web clients communicate — and securely?

TCP / UDP
Reliable delivery or low-overhead datagrams?

DEVICES
Which equipment connects the network?